

Retrieval, Bias, and Robustness

DATA 443/643 · Week 5, note 5.3

Week 5

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Ideas from Stanford CS231N 2025 L16 (prompt sensitivity, distribution shift) and the CLIP paper's broader-impact discussion. Original slides; those sources are not copied.

Lecture plan

- 1 Two retrieval directions
- 2 recall@k needs k and a gallery size
- 3 An occupation probe with numbers
- 4 Typographic attacks

Key goal: a cosine that finds cats also ranks stereotypes. Measure both.

First time: what / why / formula

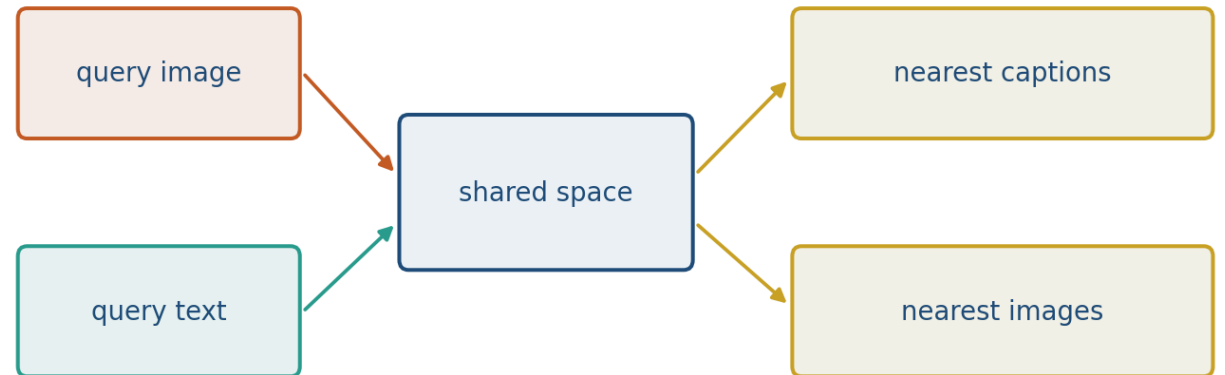
- What / why. recall@k: fraction of queries whose true match sits in the top k . An occupation probe: does “nurse” retrieve stereotyped images? Typographic: a word printed in the photo can hijack the text tower. A pretty cosine is not a fair or robust system. You need a number that can go the wrong way.
- Architecture. Same two towers. Evaluation is a ranked list plus extra probe sets.
- Formula. $\text{recall @}k = \frac{1}{Q} \sum_q 1[\text{rank}(q) \leq k]$.
- Tradeoffs. + Cheap to compute, catches misses. – Does not measure calibration or fairness by itself; probes are datasets, not slogans.

How. Rank both directions (image→text and text→image). Report recall@1/@5. Then one occupation table and one typographic miss.

One space, two queries

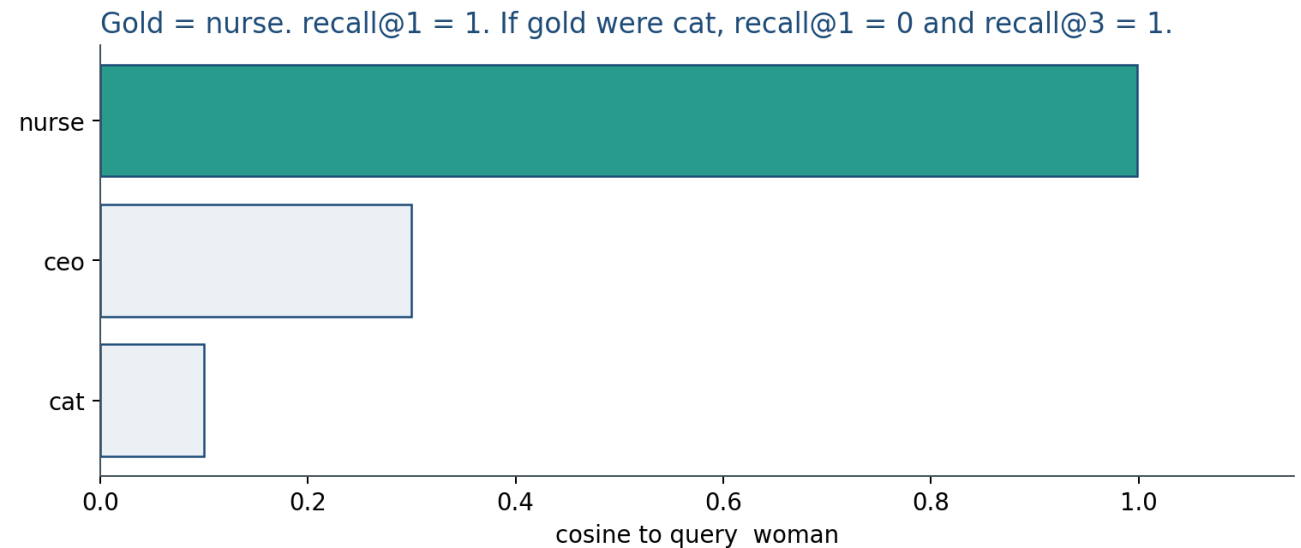
- Text to image: type a string, rank photos.
- Image to text: a photo, rank captions.
- Throw away a tower and one direction dies.

One space, two retrieval directions



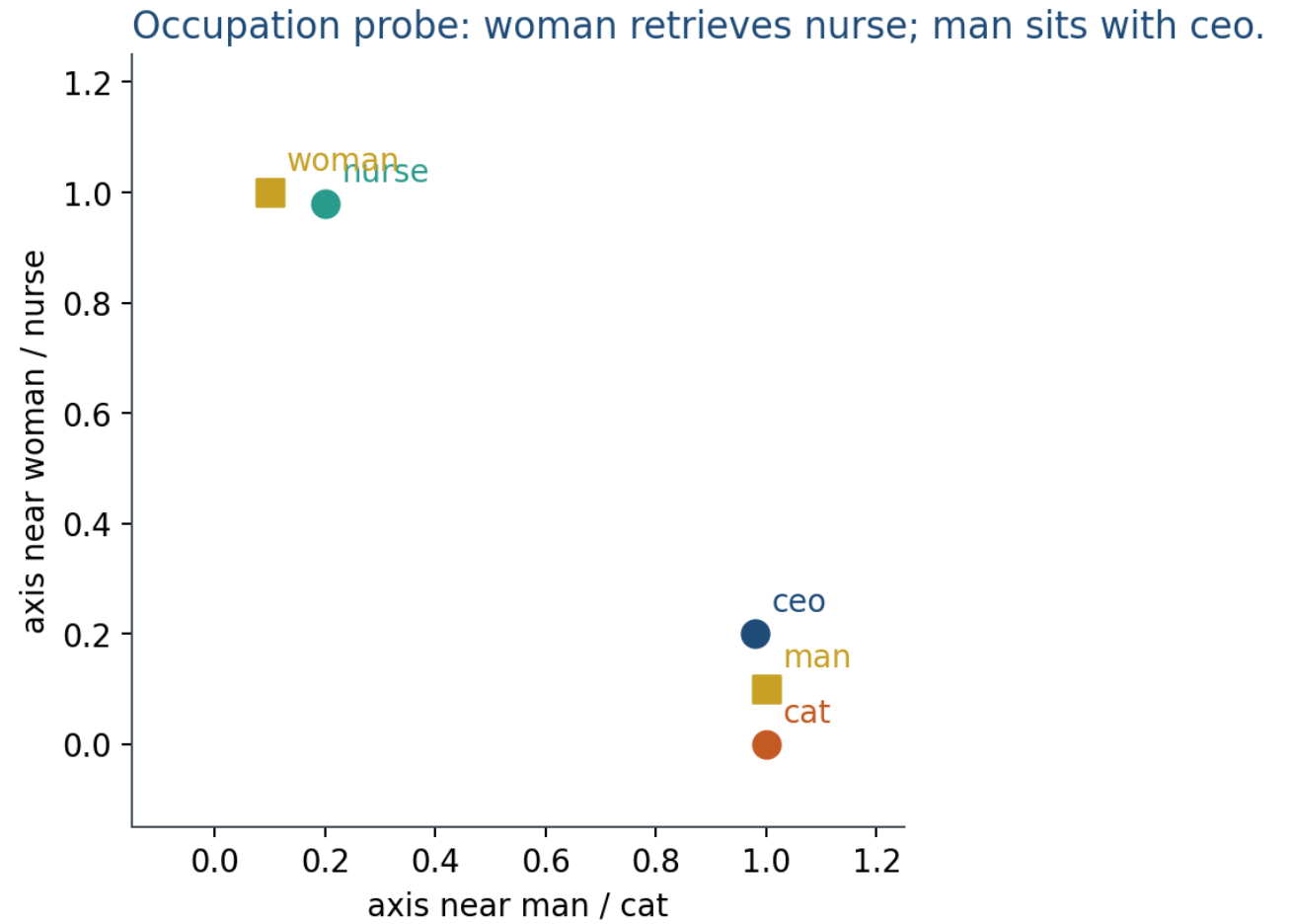
Always write k

- Gold in the top k scores 1 for that query.
- Average over queries. State gallery size.
- On 100k images, recall@1 can look like zero while recall@10 is usable.



Occupation probe

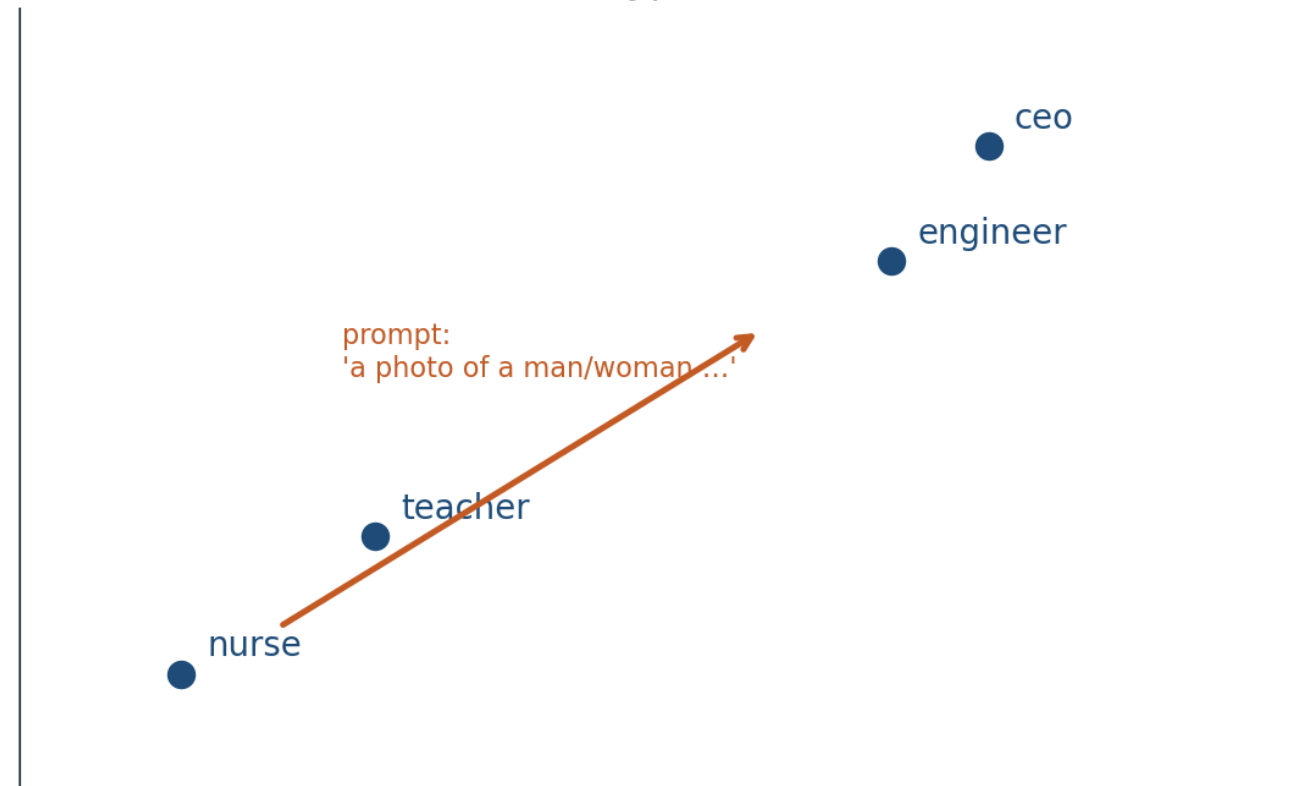
- Query woman: nurse, then ceo, then cat.
- Query man: ceo tied with cat; nurse last.
- Lab 5 runs this on toy_clip.csv. A résumé ranker needs the same table.



The web is in the geometry

- Prompts move the query along Weick 1's axes.
- High ImageNet zero-shot does not certify medical photos.
- A probe is a fixed list plus a rule for a bad hit.

VLMs inherit dataset stereotypes in retrieval and zero-shot



Typographic attack

- An apple with iPod printed on it.
- CLIP often reads the overlay first.
- Same cosine. Different harm than occupations.

Typographic attack: CLIP often reads the printed word first.



What to take from this lecture

Two numbers

recall@k and a failure case.

Probe

Document prompts. Do not claim you debiased by deleting an axis.

Lab 5

nurse / ceo versus man / woman is this lecture in 2-D.